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Overview of Advancement in Core Analysis and Its Importance in Reservoir Characterisation for Maximising Recovery

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Abstract

The expected worldwide average recovery factor is around 35% with current development strategies and practices in place. Recoveries in the offshore fields are much lower compared to onshore fields predominantly due to larger well spacing, inadequate reservoir characterisation and shorter life cycles. Efforts are in place to maximise recovery by applying Enhanced Oil Recovery (EOR) techniques. Reservoir rocks saturated with hydrocarbons are complex in both a macroscopic and microscopic scale and this complexity controls the initial quantity and distribution of hydrocarbons and flow behaviour of fluids within the reservoir. Therefore, reservoir characterisation is of utmost importance for the evaluation process in EOR. Core and log analysis along with pressure-production data greatly assist in defining the reservoir and reducing the uncertainties associated with it.

This paper discusses the importance and criticality of core analysis starting from core acquisition, preservation, laboratory studies, analysis to the application of data. Discussions are conducted for the selection of the most appropriate coring technique while lab studies were carried out on cores for reservoir description, estimation of EOR incremental oil and formation damage during injection and production processes. Critical analysis is made to highlight the quality and quantity of core analysis data needed for petrophysical interpretation, understanding the storage and flow behavior during primary, secondary and tertiary recovery stages. Important guidelines are also provided for the selection of number of plug samples for studies, laboratory methodologies, their strengths and weaknesses, and Quality Control (QC)/Quality Assurance (QA) techniques.

The paper further elaborates the recent advances in EOR processes particularly on core analysis, in-situ saturation monitoring, and the interaction between injectants and rock-fluid along with mitigation experiments. Digital core/Pore network modeling is one such emerging technique utilised for the visualisation, characterisation and special core analysis (SCAL) measurements of reservoir rocks. It can provide routine and special core analysis measurements and petrographic analysis which can be used in the quick evaluation of static and dynamic petrophysical properties and flow behavior.

Introduction

Reservoir characterisation is critical for the success of an EOR process in any field and core data can play a vital role in development drilling programmes and recovery planning operations. Well planned field coring programmes and subsequent laboratory testing procedures can strengthen log

interpretation criteria, aid in the completion/stimulation operations, provide a sound basis for reserves estimates and reservoir modeling, and supply much needed guidance in secondary and tertiary recovery programmes.

This paper is an overall review of the status of core analysis encompassing new technology in coring methods and basic laboratory testing procedures. Suggestions are also given for selecting samples for special analysis test work. Some of the objectives of the core analysis can be summarised as follows:

- defining gas-oil and oil-water contacts, formation limits, and type of production expected;
- understanding the depositional environment and stratigraphic including grain size and grain size sequences; vertical sequence of facies; sedimentary structures (ripples, cross bedding); biogenic structures (root zones, burrows); diagenetic alterations (cementing, secondary porosity, secondary mineralisation);
- understanding the areal and vertical variation of porosity, permeability, and lithology. This is required for reliable estimation of in-place volume and reserves and also for defining the mathematical models;
- defining reservoir water saturation and guidance that core should not be taken in transition zone rather should be above transition zone;
- defining reservoir net pay;
- measuring capillary pressure, wettability, relative permeabilities;
- displacement experiments with EOR injectants (Gas, Chemical, Heat) for estimating the incremental oil;
- relevant data for calibration of logs;
- estimating the magnitude of residual oil saturation and needed in enhanced oil recovery studies and utilises either a pressure or sponge core barrel);
- modified residual saturation after EOR injectants;
- hysteresis of Capillary pressure and Relative permeability;
- mineralogical studies for identifying the critical minerals important for EOR process (e.g. sensitive clays, Gypsum etc.);
- petrographic studies for studying and understanding the petro fabric of the rock;
- understanding the formation damage due to injectant and establishing the critical rates; and
- to study clay stabilisation and its mechanism.

The Value of Core Analysis

Cores provide the ground truth for fine-tuning all other sources of formation evaluation information [4]. It is used to calibrate log and geophysical data and provide plenty of detailed geological and engineering information. Unfortunately, it is relatively simple to compromise the accuracy of this data as the rock is being cut, and as it is packaged, transported, stored, and finally sampled. Fortunately, it is just as simple to maximise the accuracy of this data through thoughtful planning. Coring is an expensive proposition and would not be done if the value of the data is not expected to greatly exceed the cost of acquisition. Drilling rig time will often be the largest cost factor in a coring programme, followed by the cost of special drilling fluids. It is possible to reduce rig time by cutting longer cores. It may not be possible to influence the choice of drilling fluid, but it is crucial to anticipate the effect of mud filtrate and to plan to manage it. Core handling, preservation and transportation decisions will be made on a case-to-case basis. Rock type, oil composition, location, climate and capability are all factors in the decision process. The key is to have a plan agreed to by all stakeholders. This takes time and several rounds of discussion to finalise. Sequencing of sampling is the last issue examined. Prioritising the analytical programme from rock chips to SCAL helps ensure that samples will be available for the most valuable studies. A good plan provides a template for sampling and a timeline for results [2]. Changes are bound to occur, but the plan will reduce the risk of compromising key data sets. In summary, knowing the factors affecting the cost of core acquisition, how data priorities change over the life of a project, and how careful planning and diligent execution of those plans can maximise the value of every core.

Contribution of Core Analysis

Oil companies always depend on the quality of data from different sources for optimum field

development. Core analysis data, as well as seismic data, geological information and data from well testing, establish a fundamental basis for the understanding of a hydrocarbon reservoir storage and production potential [6]. Furthermore, the nature of different reservoirs may vary substantially, with respect to geometry, origin and content. Hence, reservoirs must be modeled individually. Nevertheless, rock properties data are frequently copied from literature or analogues because of the lack of experimental data. The motivation behind this review is to focus on the importance of proper laboratory data from core analysis for all disciplines as shown in Figure 1, which shows that core data is not only for petrophysics, but is also very important for other disciplines such as geophysics, geology, petroleum engineering, production engineering and drilling engineering. Important steps towards a comprehensive core analysis programme include core catching, rock characterisation, and selection of representative rock types, fluids and conditions. The core analysis programme itself should be designed to represent the actual reservoir challenges with respect to fluids in-place and future displacement processes. Further, implementation of resulting data into a full field model is of major importance for the design of the actual laboratory preparations and displacement processes. The data obtained from this analysis should have a great impact on minimising uncertainty for all disciplines like geology, geophysicist, petrophysicists, reservoir engineering, petroleum technology, drilling and resource assessment throughout the entire life of the field i.e. from exploration until the abundant of the field.

All the different steps from core catching and transport, to actual laboratory techniques and experimental conditions, require careful planning to obtain results with minimum uncertainties. By having proper implementation of the results from basic laboratory tests, this can give the reservoir management team vital information for further displacement and production strategy. Core analyses reveal information which are crucial for field management decisions.

Core Analysis (RCA/SCAL Analysis)

The objective of the core analysis should be established early in the development drilling programme. Common objectives include (1) definition of porosity and permeability, residual fluid saturation, lithology and predictions of possible production of gas, condensate, oil or water; (2) definition of areal changes in porosity, permeability and lithology to 3haracterize the reservoir; (3) definition of interstitial water saturation; and (4) recovery of reservoir rock in an unaltered wettability and/or saturation state for special core analysis tests to assist in defining the most favorable plan to maximise oil recovery and profitability.

Early definition of the objectives ensures that the petrophysical information required for completion or stimulations can be developed on representative core samples for assistance in subsequent wells. Correlation of measured rock properties with down hole logs will establish basic guidelines for the subsequent non-cored wells.

The team should be clear on their objectives, such as where and why they need the new core. Involvement of all disciplines in the core programme is crucial to show the benefits they will obtain from this core (reduce uncertainty). Figure 2 is a schematic detail of the routine and special core analyses test types.

What We Have in the Market Now

Generally, reservoir rock properties can be measured either on whole core samples or small core plugs that are drilled from whole core samples. The following section is a brief description regarding whole core and core plug samples as well as sidewall core samples.

Sponge Core Analysis

Sponge coring is also done to estimate the residual or remaining oil saturation at a particular stage of the producing life of the field. This provides critical data for firming up the right EOR process for the field. Full diameter analysis of samples recovered within the sponge barrel proceeds along the usual lines once the core has been removed from the barrel. The sponge itself is cut from the core barrel and the fluids it contains are extracted using a vacuum retort technique. Both oil and water

volumes within the sponge are measured. Table 1 shows residual oil saturation data for the core alone and for the core plus sponge for a specific field example. Note that the contribution of the sponge is variable and may be significant.

Whole Core

A whole core sample is basically a complete section of a conventional drilled core from a given formation. The importance of the whole core analysis lies in the fact that small scale heterogeneity (e.g. for variations in rock properties as a function of position) may not be appropriately represented in measurements on small core plug samples. The advantage of the whole core analysis is that it measures properties on a large scale, somewhat closer to that of the reservoir.

The determination of rock properties using whole core samples is, however, a much more demanding task considering the sample dimensions, large size equipments, and additional time needed to ensure control of experimental conditions. Such stabilisation, flow rates, pressure, temperature and so on, can be rather tricky. Moreover, cleaning of the whole core can also be difficult and time consuming, and laboratory analysis is generally more expensive than conventional core plug analysis.

Core Plug

A core plug sample refers to a much smaller portion of a whole core sample. A core plug sample is obtained by cutting cylindrical plugs of typically 1.0 inch or 1.5 inches in diameter and length up to 3 inches from a whole core. All necessary rock properties are typically measured on a number of such core plug samples. Generally, core plugs are cut from whole core in two different orientations, horizontal and vertical plugs.

Sidewall Cores

There is a great need for acquiring rock samples of reservoir layers as these are the means for obtaining a truly representative description of the reservoir lying several thousand feet deep.

As cutting a full core of the reservoir is expensive and must be anticipated months in advanced, several techniques have been developed to cut and retrieve core plugs from formations adjacent to the borehole**.** Percussion and rotary drilled cores are industry-standard services, which fulfill the task of performing a sidewall coring job to acquire undamaged sidewall core samples for further geological, petrophysical and geomechanical analysis, hydrocarbon show and paleontological dating. The technical aspects of such services rely on advanced computer-controlled and hydraulically-powered coring devices for cutting and retrieving sidewall cores. Normally, the coring and storage operation is continuously monitored by a graphical surface system.

Traditionally, sidewall core plugs samples have posed challenges for laboratory testing because of their diameter of 1.0 inch and length of less than 2.0 inches. Many laboratory geoscientists and engineers prefer using samples that are 1.5 inches in diameter and 2.0 inches long because it allows having more sample volume which can be of advantage to many experimental works [1].

Sidewall Rotary Cores

This is a new sidewall rotary coring technology that provides 1.5 x 2.5 inches core plug samples with a high degree of recovery (Figure 3). The samples acquired from this new technology help to provide good quality core plugs for various types of analysis (RCA/SCAL) in comparison to the old percussion one which suffered from damage and can only perform RCA [5].

Rotary sidewall coring differs from the percussion sidewall in that:

- samples are drilled from the wellbore producing a sample similar in size and quality to conventional core plugs;
- samples experience greatly reduced shattering of grains and compaction;
- sample quality is suitable for quantitative routine and SCAL;

- samples are rated for hydrostatic pressure and wellbore temperature only –no need to consider effect of explosives.

The Efficiencies of Sidewall Core Sampling

- Cost-effective alternative to conventional coring because the cores can be cut significantly faster than conventional whole core.
- Generally, the samples can be processed faster than conventional core. This makes the data available sooner for critical decisions. In fact, samples could be analysed at well site during critical situations.

The core samples can be used for a diverse set of purposes. Figure 1 schematically shows all the disciplines that require this kind of information from core analysis. The Figure 4 below shows also the areas of reservoir characterisation that are regularly supported by core data acquisition (which include sidewall cores).

Cation Exchange Capacity

The cation exchange capacity (CEC) is a measure of the ability of clay such as montmorillonite, chlorite, illite or kaolinite to exchange ions between clay surfaces and the surrounding water. It is normally expressed as milliequivalents per 100 grams of dry rock. Its magnitude is a function of the clay type and amount and may be determined using wet chemistry techniques that utilise an ammonium or a barium exchange. Cation exchange capacity is a mechanism that has been related to stripping of ions from injection fluids, trace element accumulation, and over-pressuring of the formation. These data have also been used to identify changes in environment resulting from changes in climate, sediment source, and erosion rate. A primary use of these data in the petroleum industry is in the Waxman & Smits and Waxman & Thomas equations where they are used for calculation of water saturation in clay-bearing formations. The CEC is converted to an exchange capacity per unit pore volume prior to use. Typical cation exchange capacities for various clays are shown in Table 2.

Capillary Pressure Tests

Water is retained in the reservoir pore space by capillary forces as hydrocarbons migrate and accumulate. This interstitial water, in water-wet reservoirs, adheres to sand or carbonate surfaces. Retentive forces are proportional to the water-hydrocarbon interfacial tension and the affinity of water for the rock (wetting preference), and inversely proportional to pore size. This implies that low permeability formations that are composed of very small pore spaces have high water retentive forces, hence, often contain high immobile water saturations. The measurement of capillary pressure requires that core samples be selected so that the pore radii distribution of the sample represents that of the reservoir. The data obtained in the test are used to define initial water saturation distribution in the reservoir as a function of the height above the hydrocarbon-water contact, and to furnish pore throat size and distribution data that are helpful in identifying various rock types present in the formation.

Laboratory Determination of Capillary Pressure

Three methods are generally used for determination of capillary data on rock samples:

- ***Centrifuge Method:*** In the centrifuge method, an artificial gravity using the density difference between the two fluids creates a capillary pressure gradient all along the plug and thus a saturation variation from the top to the bottom.
- ***Purcell's Method/Mercury Injection Method*:** The core plug cleaned and dried with pore volume determined is first subjected to vacuum after pore volume determination. The mercury is injected into it in increasing pressure stages. At each stage, the volume of mercury intruded is recorded. The capillary pressure is the absolute pressure of mercury.

- ***Restored State Method*:** The core plug saturated with brine is placed on a porous plate saturated with brine. Oil is injected at increasing pressure stages. A capillary tube is used to measure the volume of water expelled from the core.

All three techniques furnish multiple saturation values so as to define water saturation as a function of capillary pressure. The restored state technique has one advantage over the other two: water is present in the core samples, which allows electrical properties to be measured along with capillary pressure. The centrifugal technique is the most rapid and is the best for poorly consolidated rocks, provided they have been mounted in sleeves with screens over the sample ends. The mercury injection technique yields the maximum number of data points. It is the best one for obtaining pore throat distribution data, but the sample will be filled with mercury at the conclusion of the test and will have no further value. Higher permeability rocks have the lowest water saturation at any given capillary pressure, thus yielding a smaller transition zone. There are some suggestions, however, that this may hold true only at lower capillary pressures. The capillary pressure that yields a given water saturation is a function of the rock-wetting characteristics. Typically, this contact angle varies between the laboratory and the reservoir. Capillary pressure is also a function of the interfacial tension between the fluids in the test core at the time of testing, which differs from the reservoir value. One of the major uses of capillary pressure data is for defining the initial water saturation of the reservoir.

Pore Throat Distribution

Capillary pressure data obtained from mercury injection tests can be converted to equivalent pore radii, and *cumulative pore throat distribution for different depositional environments*. These data have been helpful in rock typing and in selection of net pay. This is an important data for selection of filter size for injection water filtration.

Relative Permeability and Hysteresis

Relative permeability is a dimensionless term that has importance when two or more fluids move through the pore spaces, for example, oil and water or gas and oil. Water-oil relative permeability can be measured on extracted cores from water-wet formations. Where the formation is believed to be intermediate or oil-wet in nature, these tests should be made on samples recovered with oil-base coring fluids (native state cores) to which no extraneous water has been added. The assumptions in this case are that the water saturation present in the sample in the laboratory is equal to that in the reservoir, that it is in the proper pore spaces, and that the wettability of the laboratory rock sample mirrors the reservoir wettability. When the formation is oil-wet, electrical properties may also need to be measured on native state core.

The terms *imbibition* and *drainage* are also employed when discussing relative permeability tests. Their meanings imply what is happening in the pore space to the wetting phase as relative permeability tests are measured. If the wetting phase is decreasing, that phase is draining and the curve is called a drainage curve. When wetting phase increases during the test, the relative permeability curve is referred to as an imbibition curve.

For a water-wet reservoir, the drainage curves apply during the time that water is draining from the reservoir and hydrocarbons are accumulating. Once the reservoir rock or laboratory sample has attained an equilibrium water-saturation value and the water is subsequently increased by natural water influx or the introduction of coring or test fluids, the imbibition curves apply. (In oil-wet rock, a reduction in the oil phase by water flooding would be referred to as a drainage curve.) These data are required in many reservoir engineering calculations, and the laboratory tests that develop them should follow the same saturation history as that in the reservoir.

Two major laboratory methods have evolved to measure relative permeability. These are referred to as the steady-state and unsteady-state techniques.

Steady State: The steady-state test, the older of the two methods, is made at low flow rates, and the test apparatus contains upstream and downstream mixer heads to remove capillary end effects. Most research groups prefer data obtained from this test. Two fluids are injected simultaneously into a core sample and the water saturation is increased slowly. This simulates the slow increase in water saturation that would occur in the formation between the injection and producing wells. Saturation increase is monitored by measuring the gain in weight occurring in the sample or by X-ray technique.

Unsteady State: The unsteady-state technique uses viscous oil and is normally made at a higher flow rate than that present in the reservoir. It is this higher rate that sometimes yields pessimistic estimates of recovery from rocks of intermediate wettability.

The natural preference of a porous medium, which causes one fluid to adhere to its surfaces rather than another, is referred to as wettability. A water-wet porous medium causes water to adhere to its surfaces. The wettability of a rock has a dramatic influence on relative permeability curves. It is therefore necessary that the core samples tested in the laboratory reflect the actual formation wettability, and that initial water saturation in the test sample be of the same magnitude and have the same spatial location as it has in the reservoir. This need has led to the recovery of "native state" cores. These are cores taken with crude oil or with other oil-base fluids that do not alter the wettability or water saturation present in the recovered core. These data indicate that as the rock becomes more oil-wet, the relative permeability to oil decreases and the relative permeability to water increases at any given saturation. This results in an unfavorable recovery efficiency. It also indicates that the residual oil saturation in intermediate to oil-wet rocks is a function of the volume of water that flows through the core sample, and that the relative permeability to water existing at flood out will be much higher for the oil-wet formation.

Immiscible Water Alternating Gas (IWAG) Lab Experiment

IWAG experiments were designed on composite core to study the mobilisation of oil that are normally bypassed in gas and water flood alone. The IWAG procedure is shown in Figure 5. For the repeated IWAG laboratory experiments, material balance analysis yielded a similar final oil recovery as that determined by volumetric measurements. The material balance showed an IWAG final oil recovery of 60.1% pore volume or 73% original-il-in-place (OOIP) compared to an oil recovery of 60.9% pore volume derived from volumetric measurement. The volumetric value is slightly different because it was measured using test tube while material balance used dean stark analysis. The composite core yielded a recovery of 43.6% pore volume (PV) or 53% OOIP. Note that from the recovery curve, it can be seen that further waterflooding would recover more oil. The injection fluid was then switched to synthetic gas, an incremental oil recovery of 6.6% PV or 8% OOIP was observed after 2 PV of gas had been injected, demonstrating a total oil recovery of 50.2% PV or 60.98% OOIP at this displacement stage. The composite core was then flooded with brine again. After 2 PV of brine injection, the total oil recovery reached 59.2% PV or 71.9% OOIP, indicating an increase in oil recovery of 11% OOIP compared to the recovery after the first gas injection. This happened because at that particular time, most of the big pores were occupied by the gas phase and when the brine was introduced into the core, a fraction of the gas was trapped inside the big pores and as a result, the injection brine was forced to flow through small pores which were bypassed previously and recovered more oil. Subsequent gas and water flooding gave little oil production.

The second WAG test showed a comparable oil recovery between the material balance check and the volumetric measurement. IWAG oil recovery from the material balance is 60.1% by pore volume while the oil recovery derived from the volumetric measurement is 60.9% by pore volume.

- First Water Flood (2 PV) = 53 % OIP (43.6 % PV)
- Incremental due to first cycle of WAG (inject Gas 2PV followed by water 2 PV) = 18.9 % OIP (15.5 % PV)
- Incremental due to second cycle of WAG = 1.45 % OIP (1.19 % PV)
- Incremental due to Hysteresis Effect = 7.41% OIP (6.1% PV)

Formation Damage by IWAG

Formation damage by IWAG injection was conducted in order to determine the effect of water gas interface on particulate movement and fluid-fluid interaction during the IWAG cycles. The test was conducted by using seawater and methane. Methane was selected as substitute gas because the planned injected hydrocarbon gas contains 89% of methane. The effect of other components on particulate movement is not considered significant enough to undertake the high cost of preparing the synthetic hydrocarbon gas for this test.

The test was conducted with field equivalent injection rates, 20cc/min seawater injection alternate with 100cc/min methane injection. The seawater filtered size used is 2 microns. The result for two water-gas cycle is shown in Figure 6.

The permeability decline during the first water cycle was similar with the previous water flood test that was conducted. During the first cycle of gas injection, the permeability was stabilised to about the initial permeability of end of the first water cycle. It can be concluded that no damage was observed from the first water-gas cycle. The second water cycle did not reach stabilisation but the final water cycle did and indicates further decline in permeability. Decline rate or pressure decline for the first water cycle is 4.0328 psi/hr; the second cycle is 4.8745 psi/hr; and the third water cycle is 4.1643 psi/hr while decline rate for the gas injection is 1.25 psi/hr. A micro-CT scan was carried out to understand the fines migration/grains movement at pore level. In Figure 6, it was observed that, in high resolution images, >90% of the sample volume shows some changes occurring in the finer pore spaces; movement of fines during formation damage (FD) with no indication of pore throat blockage noted. It was observed that field potential damage can be classified into four main formation damage mechanisms which are fines migration, particulate damage, clay swelling and inorganic scaling. Critical rate of injection was established based on formation damage study. Typical half life for two permeability system is shown in Figure 7.

Unconventional Resources

One key element in unconventional plays is the collection of sufficient data to enable accurate core-to-log calibration. In this technique, traces are derived to enable characterisation of bulk rock properties (mineralogy, organic content, porosity, and water saturations) and total (adsorbed, absorbed and free gas) gas-in-place (GIP) volumes over the entire logged interval (at the same resolution of the log digits) from a discrete set of core data. Core-to-log calibration is critical for shale gas reservoirs because the standard petrophysical modeling techniques are problematic or irrelevant in shale formations. The predominat complications are: the fact that the adsorbed GIP (frequently as much as 50% of total gas) has very little impact on log response; water resistivity is unknown (shale formations do not produce water); and the Archie Equation is not applicable due to the lack of clay-corrected parameters.

Based on the resulting core-to-log evaluations, informed decisions are made possible to determine the economic potential and net zones of pay. If indeed the play has potential, further analyses are then justified to characterise the reservoir more comprehensively and help facilitate appropriate completion strategies.

The Future of Core Analysis (Digital Core/Pore Network Modeling (PNM))

Digital core is a term used to describe a more detailed analysis of 2D/3D CT scan images. First, higher resolution images than conventional CT scans are made. Then the pore size and shape are quantified in three dimensional spaces, and a series of well-defined equations are used to generate computed rock properties such as porosity, permeability, capillary pressure, relative permeability, electrical properties, mineralogy, elastic properties, and so on. Some of these can be defined in the three primary directions (X, Y, and Z directions). Total, effective, connected, unconnected and fracture porosity can be derived. Permeability as low as a few nanoDarcies can be detected [3]. Samples can be taken from cores, core plugs or drill cuttings. The latter allows measurement of quantitative rock properties without coring - a significant cost saving on current wells and an invaluable source of new information on older wells.

Pore scale modeling offers exciting possibilities for bridging the gap between detailed descriptions of the variability of single phase properties and the lack thereof for multiphase flow properties [8]. Recent advances in pore scale modeling have shown that a realistic characterisation of the pore structure can be used to produce a model that accurately predicts both single and multiphase flow properties, at least for diagentically simple rocks. If predictive pore scale modeling is verified for a diverse range of reservoir rocks, it can be a useful reservoir characterisation tool which can be used to populate fine scale geological models with a realistic spatial distribution of constitutive relations derived from pore scale modeling Figure 8.

It is well established that small-scale heterogeneities can have a significant impact on flow and oil recovery. However, small-scale heterogeneities cannot be directly incorporated in field-scale simulation models because of limitations in computer speed. Therefore, the average effects of small-scale heterogeneities in large-scale numerical grids must be accounted for through upscaling [7].

The main purpose of this part is to introduce an integrated pore-to-field scale upscaling procedure for multi-phase flow in heterogeneous reservoirs Figure 8.

Digital Core Analysis

Digital core analysis was carried out on the rock samples to understand the potential changes of pore morphology, fluid distribution and wettability during the handling and execution of lab experiments (Figure 9). Evaluation of the impact of heterogeneity and wettability changes on petrophysical properties for drainage and imbibition effects. Routine and special core analysis data were also generated. Grain size distribution and pore size distribution was also studied, the data of which can be used for design of filter. Micro CT images were generated for sandstone, heterolithic and mudstones to understand the pore morphology. Field Emission Scanning Electron Microscopy (FESEM) images were acquired for the investigation of wettability (Figure 10). United state bearuro of mines (USBM) index is W=-0.052 and Amott Harvey wettability index is Iw=-0.046. This shows that the rock is intermediate wet. The RCA results of porosity and permeability measurements measured on core plugs show the same distribution and trend to the eight core plugs with MCT derived PNM properties. This is shown in (Figure 11) and indicate a good match on the porosity-permeability cross plot. A sensitivity analysis was carried out to illustrate the effect of wettability, flooding rate and fluid type on the prediction of capillary pressure and gas/oil relative permeability curves. Comparison of the relative permeability for the clean sand subsets of heterolithic sample and laminated sample under water wet (WW) and mixed wet (MW) conditions were used for simulation. Kr plots in Figure 12 shows the clear difference in the relative permeability curves. This variation is due to local contact angle (+/- 10 degree) and wettability conditions (45-60% WW surface) while preserving the Amott Index=0. Relative permeability for laminated and heterolithic samples are shown in Figure 12.

Conclusion

The planning and the objectives of the coring must be clear and on purpose, which help to select the kind of coring technology to be used. Where whole core has not, or cannot be obtained, sidewall cores have proven to be a viable alternative. Drilled sidewall cores can contribute to improved reservoir characterisation as well as an understanding of formation heterogeneity when the tops and ends of individual cores can be recognised. Most of the RCA/SCAL can be carried out on the rotary core samples. With the minimum volume of the core sample, we can carry out core analysis using digital core even on a small part of the core or even drilling cutting. Most rock properties can be measured now using digital core and can be upscaled from micro-scale to field-scale. Digital core analysis can play a vital role in unconventional plays such as tight gas, oil shale, gas shale and CBM where it can facililate some analysis to characterise and model the type of reservoir.

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Table 1: *Core, sponge, and core-plus-sponge residual oil saturation data*

Depth ***Feet***	**Core residual oil** ***% pore space***	**Sponge residual oil** ***% pore space***	**Total (sponge plus core) residual oil** ***% pore space***
4636	23.1	0.8	23.9
4637	23.1	0.8	23.9
4638	21.7	8.3	30.0
4639	20.4	10.0	30.4
4640	28.7	7.0	35.7
4641	22.1	6.2	28.3

Table 2: *Cation exchange capacities (CEC) of various clays*

Clay	CEC (MEQ/100 grams)*
Montmorillonite	80-150
Illite	10-40
Chlorite	10-40**
Kaolinite	3-15

*Units are milliequivalents/100 grams of dry rock.

**In some studies reported the CEC has been as low as 0.0 to 1.0

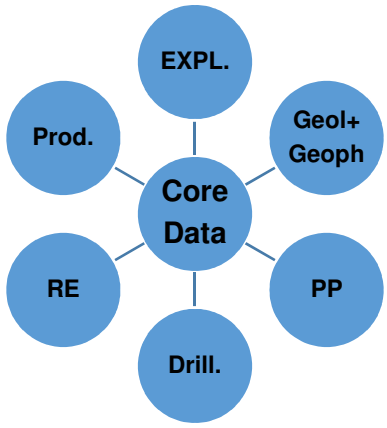


Figure 1: Contribution of core analysis to all disciplines.

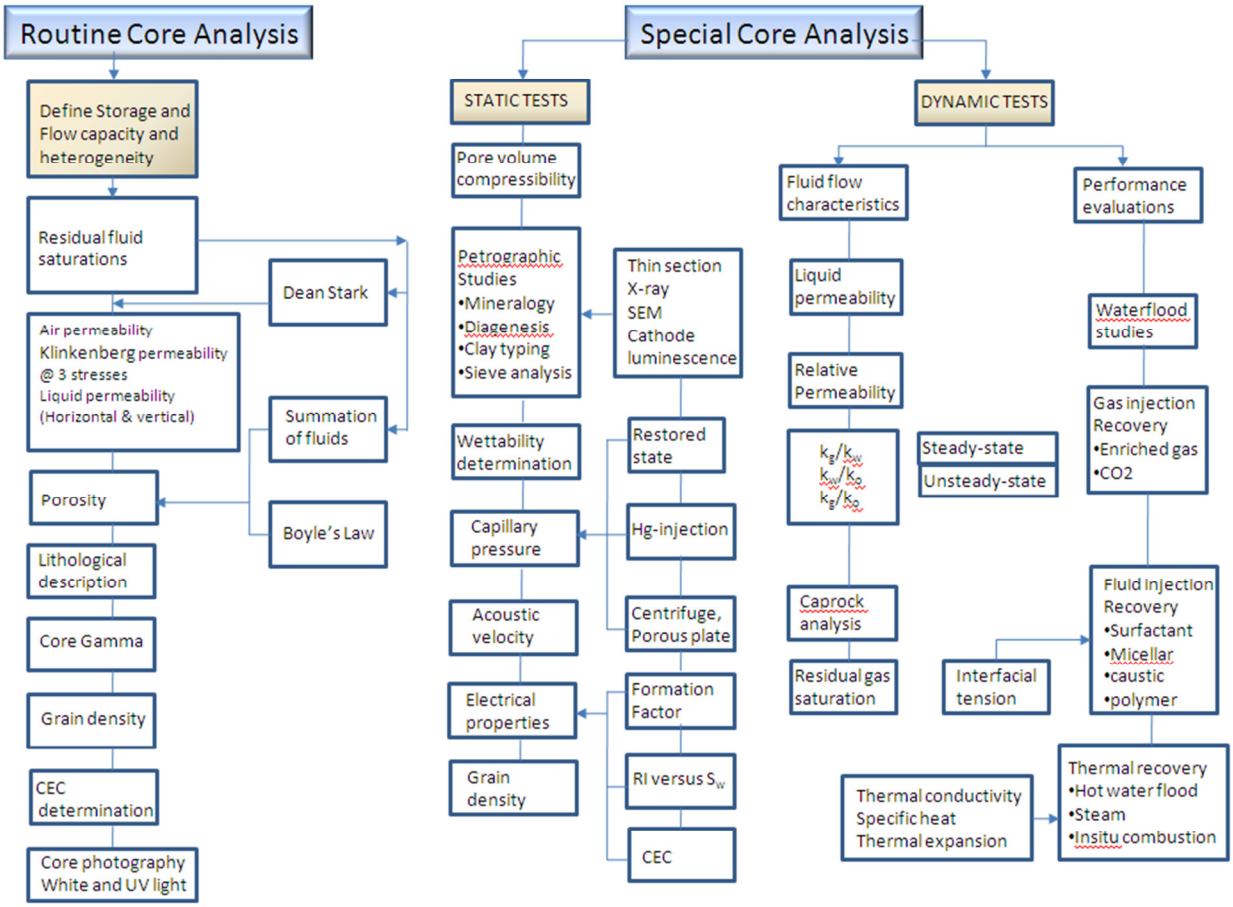


Figure 2: RCA and SCAL Workflow.

Figure 3: Bit section of standard rotary sidewall coring tool and retrieved core plug samples.

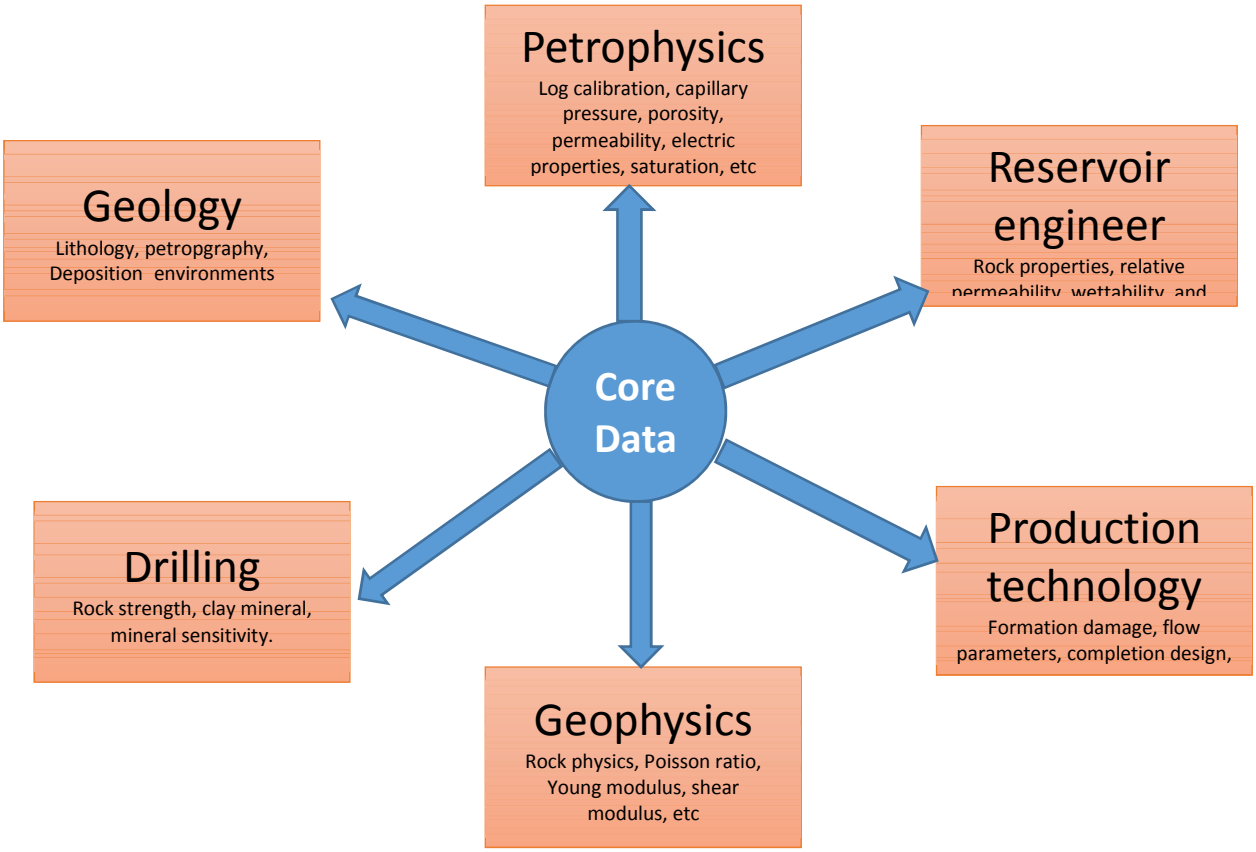


Figure 4: Areas of subsurface characterisation supported by core data.

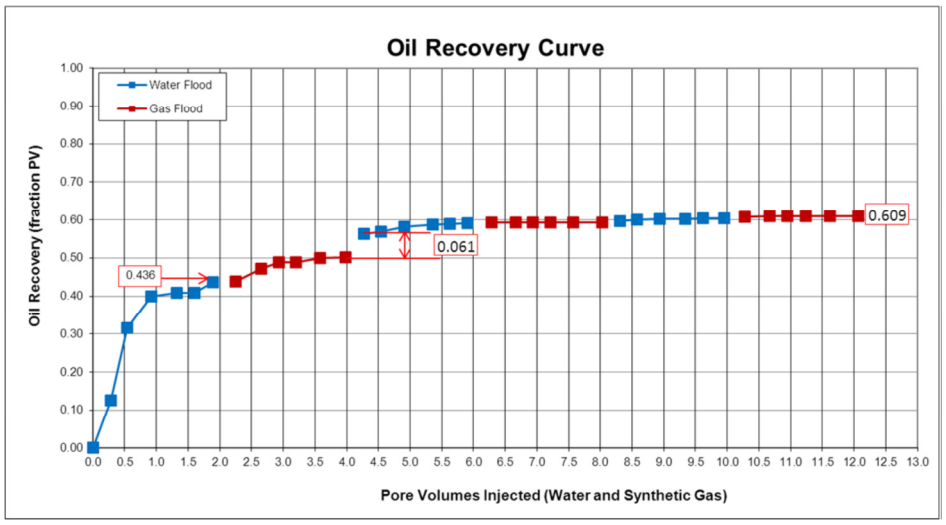


Figure 5: IWAG Lab Experiment.

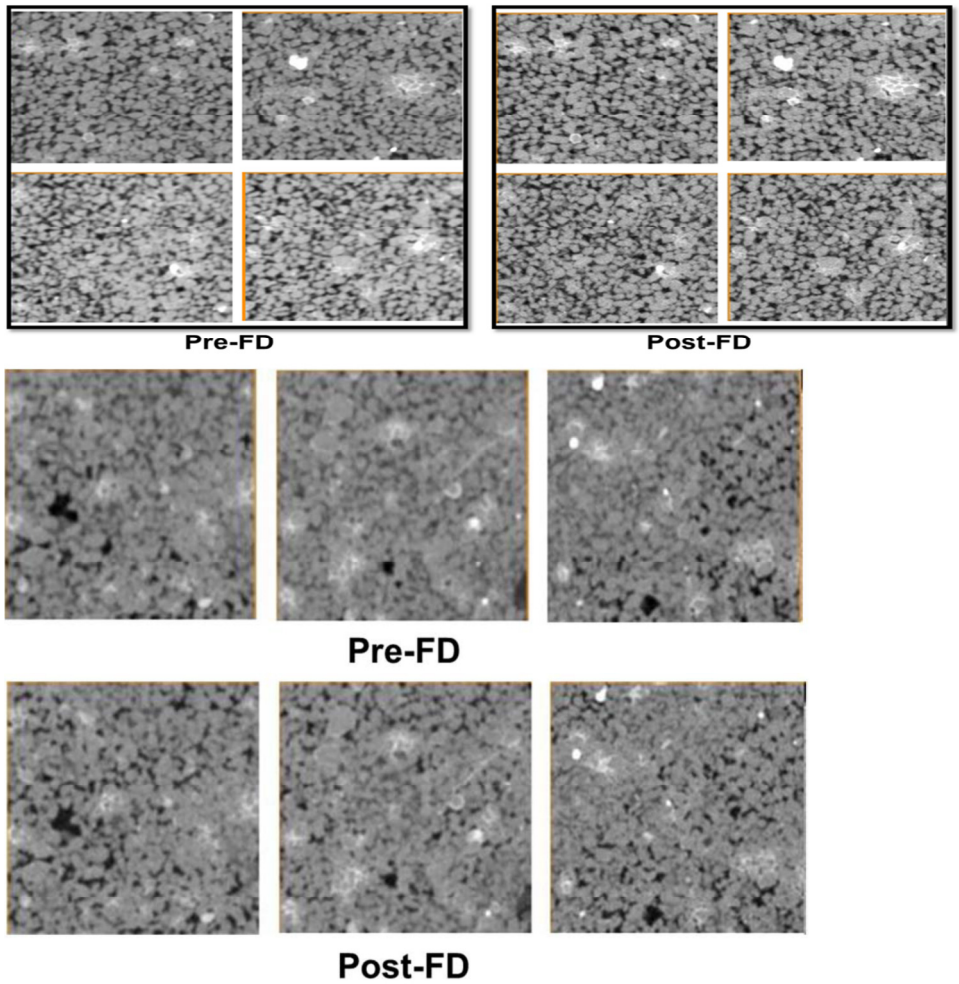


Figure 6: Formation Damage Pre- and Post-CT Scan.

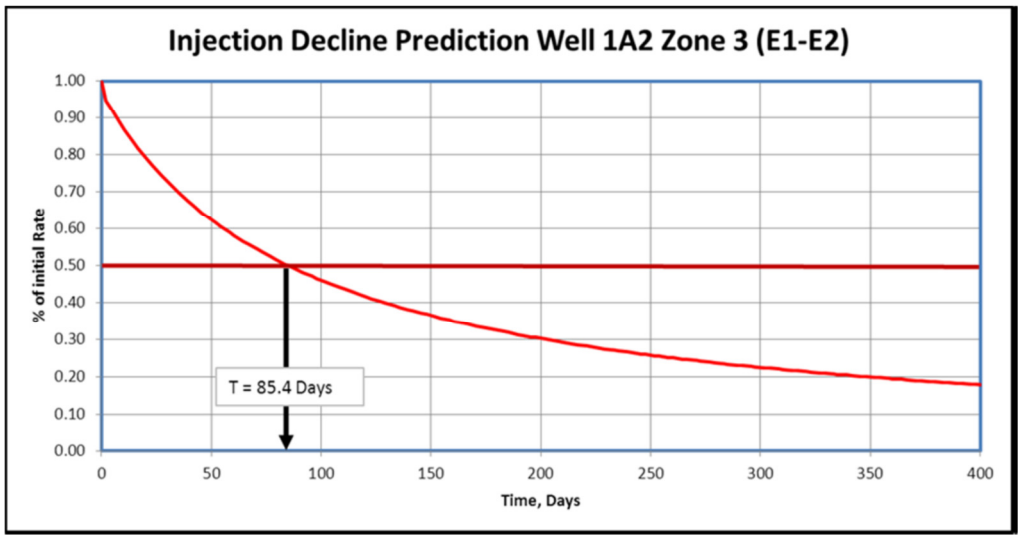


Figure 7: Half Life Injector

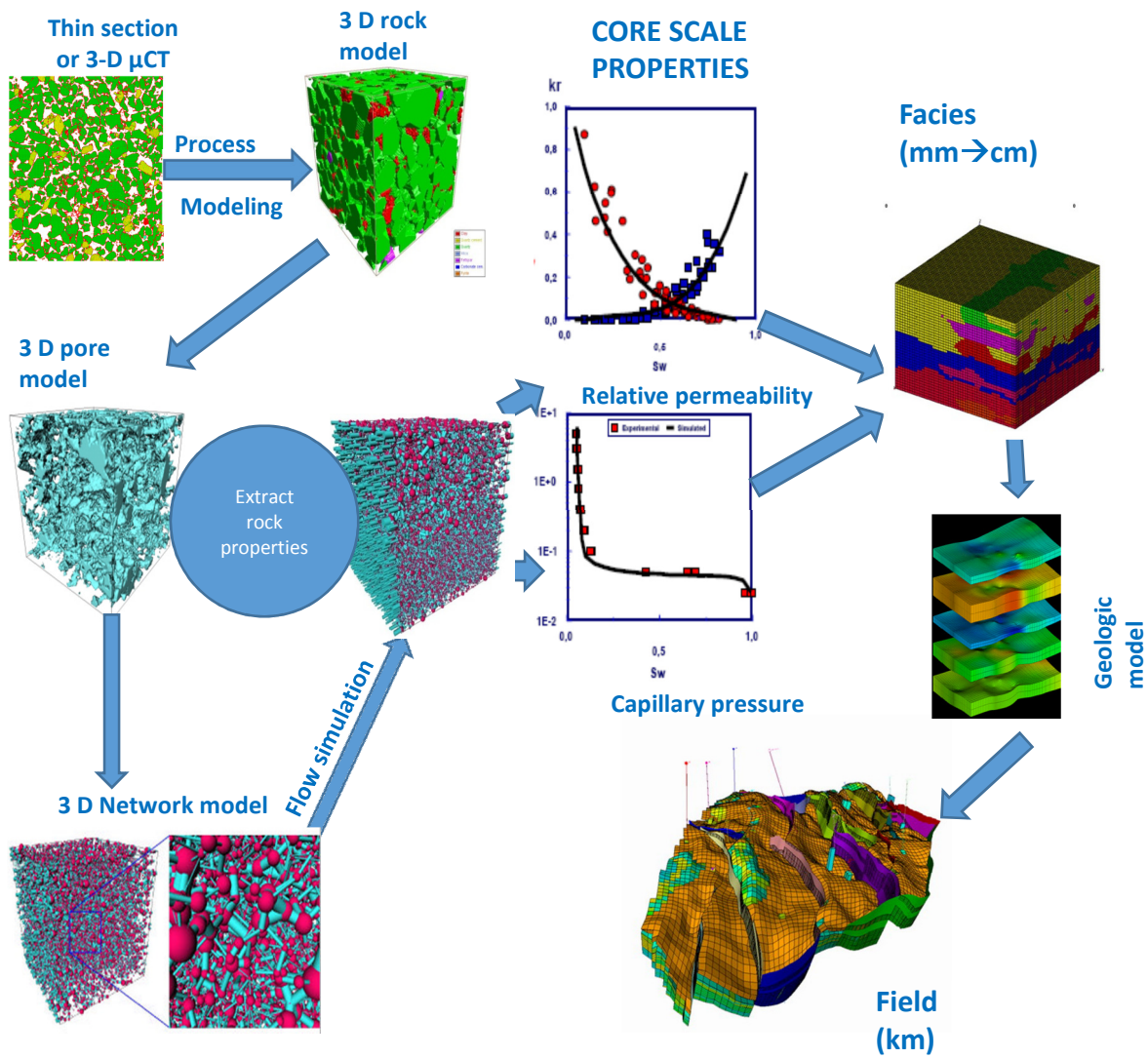


Figure 8: Illustration of the major steps involved PNM workflow from pore to field scale.

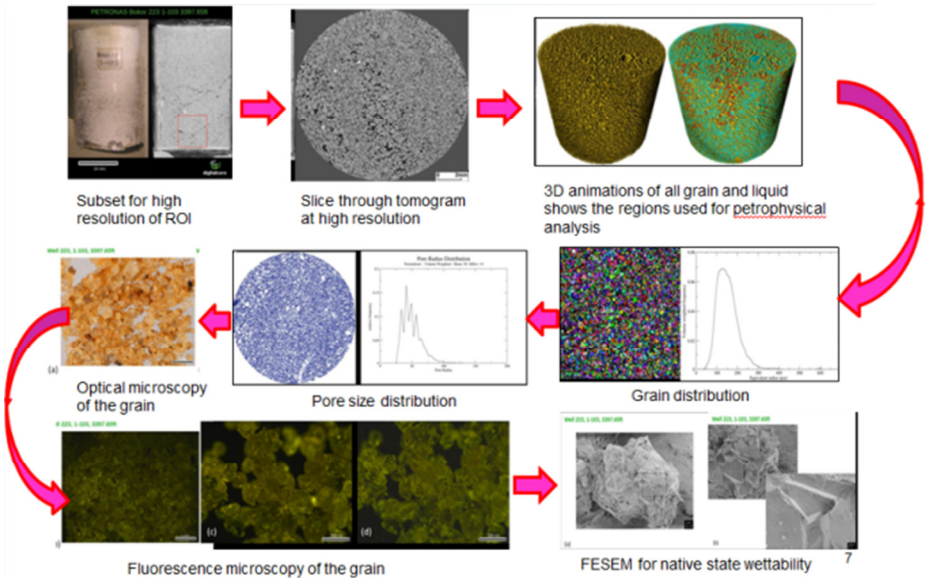
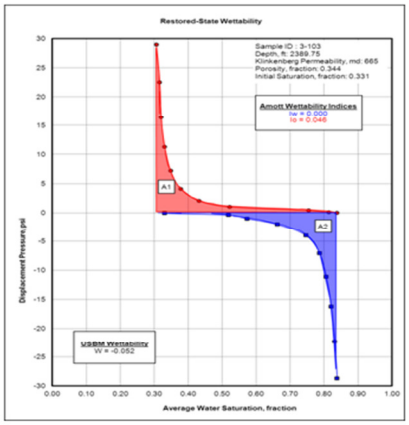


Figure 9: Digital Core analysis workflow.

Approximate relationship between wettability. Contact Angle, USBM and Amott Indexes (Anderson, 1986)

	Water-wet	Neutral Wet	Oil-wet
Contact angle			
Minimum	0°	60° to 75°	105° to 120°
Maximum	60° to 75°	105° to 120°	180°
USBM wettability index	W near 1	W near 0	W near -1
Amott wettability index			
Displacement-by water ratio	Positive	Zero	Zero
Displacement-by oil ratio	Zero	Zero	Positive
Amott-Harvey Wettability Index	0.3 ≤ I ≤ 1.0	-0.3 < I < 0.3	-1.0 ≤ I ≤ -0.3

USBM Wettability	
Area 1	**Area 2**
1.229	1.386
Wettability Number	
W =	-0.052
Amott Indexes	
Iw = 0	Io = 0.046



Apparent Wettability (as tested):
Neutral/Intermediate Wet

Figure 10: Apparant wettability, realtion between wettability, contact angle, USBM and Amott indexes.

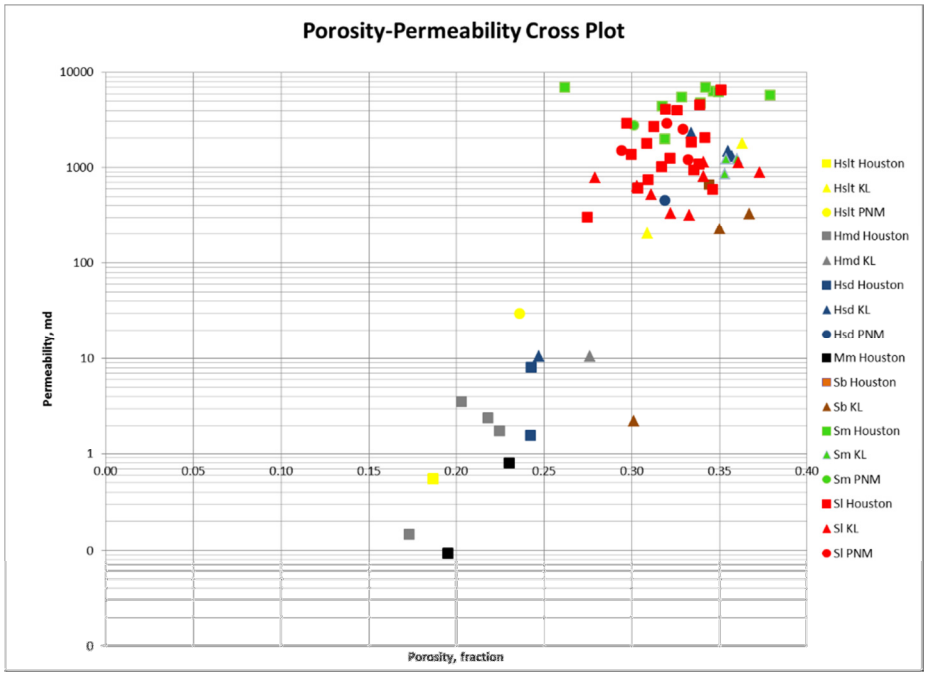


Figure 11: Porosity-Permeability Cross Plot.

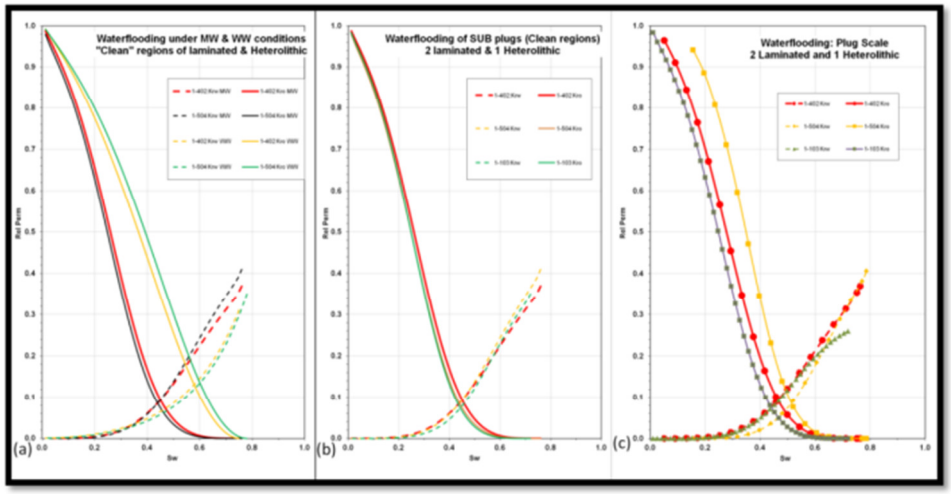


Figure 12: Relative Permeability Curves.

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